

HelpMate — Prototype Overview & User Guide

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Course: INFO90010 Technology Innovation Project

Team: Group B · University of Melbourne, 2026

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Important Links

	Link
Repository	GitHub
Technology Demo Video	Demo Video
Final Pitch Video	Pitch Video

1. Access Instructions

Dev Account (Recommended for Quick Testing)

A pre-configured developer account is available with existing posts, exchange history, reviews, and leaderboard data already set up — no setup required.

Display Name: Alex Chen

Note: On the login screen, tap the ⚡ **Dev: Login as Alex Chen** button at the bottom for one-tap access.

Alternatively, register a new account with any email and password via the Sign Up screen.

Platform Requirements

Tool	Version Required
Node.js	v18 or later
npm	v9 or later
Expo Go app	Latest (App Store / Google Play)

2. Testing Instructions

Option 1 — Expo Go on Mobile (Recommended)

Works on any iPhone or Android phone. No Xcode or cable required.

1. Clone the repository and enter the project folder:
<https://github.com/Minion1435627/Technology-Innovation-Project.git> && cd HelpMate
2. Install JavaScript dependencies: npm install
3. Install the tunnel package: npm install -g @expo/ngrok (optional)
4. Start the server: npx expo start
5. Scan the QR code — iPhone: use the Camera app; Android: use Expo Go → Scan QR code

6. Log in with the dev account above, or create a new account

Option 2 and Option 3 (Link)

Features to Test (Link: For all features details)

Screen	What to Try
Map	Tap the Need / Supply / Friend filter pills; tap a marker to view the tooltip
Post	Create a Need or Supply post using the + button on the map
Chat	Start an exchange from chat;
Leaderboard	Filter by area or task category; observe the bar chart scores
Profile	View reviews, level badge, exchange history
Mini-games	Open Farmer or Fisher games

Known Limitations

Limitation	Detail
Push notifications	Not yet implemented. app must be open to receive messages in real time
Identity verification	Verification badge is UI only. No backend check performed
Google OAuth	Login button present but native redirect not fully configured
Avatar in Simulator	3D avatar may not render in the iOS Simulator; use a real mobile device
Score penalties	Overdue/dispute deductions are designed but not yet wired to the backend

3. Core Features & User Needs Addressed

Feature 1 — Map-Based Help Discovery

What It Does:

When a user opens HelpMate, they land on a live neighbourhood map showing all nearby help requests (red markers) and supply offers (green markers), with friends shown in blue. Each marker displays the poster’s two-letter initials. Users can toggle the **Need / Supply / Friend** filter pills below the search bar with a single tap. Tapping any marker opens a detail card showing the post, the poster’s rating, verification badge, and a Contact button.

User Needs Addressed:

User Need	How This Feature Addresses It
No fast, reliable way to find nearby peer help in real time	Live map shows all nearby posts with distance and urgency at a glance
Having to use multiple fragmented platforms	All discovery happens in one screen — no switching between apps
Fear of interacting with unverified strangers	Tooltip shows the poster’s rating, badge, and level before any contact is made

Feature 2 — Post & Exchange System

What It Does:

Users create a **Need** (requesting help) or **Supply** (offering help) post with a title, description, category, urgency

level, expiry time, and optional photos. When a user taps **Contact**, an in-app chat opens with the post pinned at the top. Either party can start a formal exchange, which moves through a tracked lifecycle:

pending → in_progress → completed / overdue / disputed

After completion, both parties are prompted to leave a mutual review.

User Needs Addressed:

User Need	How This Feature Addresses It
No platform designed for everyday micro-help tasks	The post system is built for small, local, time-limited help requests
High ghosting and no-show rates	The tracked lifecycle with countdown timer and lockout creates mutual accountability
Unnecessary purchases when borrowing would suffice	Borrowing and lending are first-class post types

Feature 3 — Trust, Reputation & Gamification

What It Does:

Every user has a public profile showing their star average, written reviews, completed exchange count, and level badge. A **weekly leaderboard** ranks users by helpfulness score with a horizontal bar chart per row showing scores relative to the top scorer. Users earn **XP** by completing daily and weekly tasks, driving a 5-level progression (Newcomer → Helper → Trusted Neighbour → Community Pillar → Legend). Two casual mini-games: **Farmer** and **Fisher**, are tied to the user's real-world level and give users a reason to stay active even before a critical mass of nearby posts exists.

User Needs Addressed:

User Need	How This Feature Addresses It
No track record visible — every first interaction is a blind date	Every profile shows star average, written reviews, and completed exchanges before contact
Helpers receive no recognition for giving	The leaderboard, XP system, and level badges make helpfulness publicly visible and rewarded
Cold-start problem — low engagement before critical mass	Mini-games give users a reason to open the app regularly before the community grows

For full feature documentation see [All_Features_Design.md](#) and [Core_Features.md](#) in the repository.